

SENTINELWELL AI

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Predictive Personnel Stress & Welfare Monitoring System for Uniformed Forces

Ministry of Home Affairs (MHA) / CRPF
Maulana Azad National Institute of Technology (MANIT),
Bhopal

OPERATIONAL MANDATE: Transform workforce mental health and fatigue management across CAPFs, Armed Forces, and Police from a *reactive crisis-response mechanism* to an *automated, explainable, and proactive welfare framework*—without psychiatric diagnostic labeling or career penalties.

14–21 Days

EARLY DETECTION WINDOW

< 85 ms

DUAL-MODEL INFERENCE
LATENCY

100% XAI

SHAP ROOT-CAUSE ATTRIBUTION

Zero Stigma

ANONYMIZED COMMANDER
VIEW

1. OPERATIONAL PROBLEM STATEMENT & DEFICIENCIES

Personnel serving in Central Armed Police Forces (CRPF, BSF, CISF, ITBP, SSB), the Armed Forces, and State Police formations routinely operate in hazardous environments involving prolonged remote deployments, irregular circadian rhythms, frequent night watches, and family separation. Today's welfare frameworks suffer from three severe limitations:

Latency in Identification

Stress is observed manually only after operational breakdown, behavioral decline, or critical incidents occur.

Career Stigmatization & Fear

Troops hesitate to report psychological strain due to fears of peer judgment, unfitness categorization, or adverse postings.

2. THE SENTINELWELL TECHNOLOGICAL APPROACH

SentinelWell AI introduces an indigenous intelligence pipeline fusing objective duty roster telemetry with voluntary, low-friction self-check-ins:

- Dual-Model Machine Learning Engine:** Couples an unsupervised *Isolation Forest* (detecting abrupt schedule anomalies) with a *LightGBM Gradient Boosted Classifier* (classifying risk across LOW, MODERATE, HIGH, and CRITICAL bands).
- Explainable AI via SHAP Game Theory:** Eliminates algorithmic black-box opacity by extracting exact Shapley values to pinpoint the top three primary trigger stressors (e.g., *14 consecutive duty days, 8 night shifts in 30 days, < 5h sleep*).
- Support-Oriented Action Hub:** Replaces punitive measures with proactive organizational solutions: mandatory 48-hour rest cycles, roster rotations, accelerated leave coordination, and voluntary confidential counseling.

3. STRICT ROLE-BASED AIR-GAPPING & PRIVACY GOVERNANCE

Portal Level	Operational Role & Access Boundaries	Privacy & Security Controls
Personnel Portal (ID: P-1024)	Voluntary sub-minute check-in (Stress, Sleep, Energy, Workload sliders), 6-month historical self-trajectory, and confidential direct welfare requests.	Granular opt-in/opt-out toggles for all wellness inputs. Biometrics strictly disabled by default.
Welfare Officer (ID: WO-208)	Prioritized triage command center (165+ personnel), 0–100 risk radar, SHAP trigger breakdown, and intervention logging.	Strictly confidential one-on-one profiles air-gapped from administrative rosters.
Commander Overview (ID: CO-014)	Battalion readiness heatmaps, unit-by-unit fatigue curves, continuous deployment distribution, and leave utilization trends.	Strictly Anonymized: Zero access to individual medical or personal identities.
Security Admin (ID: AD-001)	System provisioning, real-time stress anomaly simulation engine, and immutable security audit trails.	Every access query is cryptographically logged with timestamps and user roles.

4. ENGINEERING STACK & CLOUD INFRASTRUCTURE

Engineered with production-grade microservices: **React 19 + TypeScript + TanStack Router** on edge nodes, **Node.js + Express API** gateway with Prisma ORM, **FastAPI + Python 3.11** for model inference, and **Aiven Cloud MySQL** over enforced TLS/SSL encryption with automated failover.

5. PROJECT TEAM — MANIT BHOPAL

Team Member	Designation & Core Responsibilities
Puru Yadav	Team Leader · AI / ML & Predictive Analytics Lead
Nishant Pastor	Frontend Architecture & UI/UX Specialist
Raghav Jhalani	Data Engineering & Pipeline Architecture
HarshVardhan Khare	Model Optimization & Machine Learning Benchmarking
Aarti Misra	Quality Assurance & Defense Domain Research
Sarthak Mittal	Backend Architecture & Cloud Deployment